

NQS-Updates

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1 Background Theory

Reference: [arXiv:1703.03152v2](#). Equation numbers below refer to that paper.

1.1 TF Ising model

After rewriting the spin operators as fermions using the Jordan–Wigner transformation (Eq. 15), we get a free-fermion (Gaussian) Hamiltonian, i.e. one that is quadratic in the fermionic operators. Everything then gets captured by the covariance matrix

$$M_{ab} = \frac{i}{2} \langle [m_a, m_b] \rangle, \quad (1)$$

a real antisymmetric $2L \times 2L$ matrix built from the Majorana operators m_a . Since the quenched Hamiltonian is still quadratic, the state stays Gaussian at all times and the covariance matrix only rotates, $M(t) = Q(t) M(0) Q(t)^\top$ with $Q(t)$ orthogonal.

A sudden quench is a simple way of driving the system out of equilibrium:

1. Prepare the ground state of an easy Hamiltonian, where $J = 0$. With only the field term left, this ground state is the fully polarised product state.
2. Quench at $t = 0$ by switching the Hamiltonian into critical mode, $J = B$.
3. Evolve the state, up to $t = L/8$ in the paper.

To clarify.

- What is the string length?
- Importance sampling in Hilbert space vs. mode space. How is it connected to Fig. 2, top right?

Derivation of the Trotter error

The true evolution can be written as:

$$U = e^{-itH} = (e^{-i\Delta t H})^T, \quad (2)$$

with Δt the step duration and T the number of steps, so that $\Delta t = t/T$. This splitting is exact because H commutes with itself.

The device evolution instead alternates the two parts of the Hamiltonian:

$$U_T = \left(e^{-i\Delta t H_B} e^{-i\Delta t H_J} \right)^T. \quad (3)$$

Expanding a single step of each to second order and subtracting, the leading mismatch is a commutator,

$$e^{-i\Delta t H_B} e^{-i\Delta t H_J} - e^{-i\Delta t (H_B + H_J)} = -\frac{1}{2} [H_B, H_J] \Delta t^2 + \mathcal{O}(\Delta t^3), \quad (4)$$

so writing $C = \frac{1}{2} \|[H_B, H_J]\|$ for the constant prefactor, a single step is wrong by at most $C \Delta t^2$.

After T steps the total error is

$$T C \Delta t^2 = C \frac{t^2}{T}, \quad (5)$$

since the errors of the individual steps add rather than multiply (each factor is unitary, so an existing error is carried forward without being amplified).

The connection to fidelity:

$$1 - F \leq \left(C \frac{t^2}{T} \right)^2. \quad (6)$$